

Revised Generation Interconnection System Impact Study Report

For Queue Project Q13-0247

Dooville 230 kV
December 25, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0247. The Interconnection Customer proposes a 252.0 MW natural gas combined cycle facility interconnecting at the Dooville 230 kV substation in the Indiana area. The request was received on June 21, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 169 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 14 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$22,713,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Dooville – Montpelier Electric Generating Station Substation 230 kV
Contingency: Wabash – Montpelier Electric Generating Station 500 kV
Loading: 119.8% (Rating: 562 MVA)
DFAX: 53.7%
Pre-project loading: 95.7%
2. Facility: Ritter Park – Sugar Hill 345 kV line
Contingency: Hilltop – Piney Crk Project 500 kV
Loading: 112.8% (Rating: 1,250 MVA)
DFAX: 11.4%
Pre-project loading: 110.5%
3. Facility: Medora – P.T.S 345 kV
Contingency: Sugar Hill – Ritter Park 345 kV
Loading: 111.9% (Rating: 1,258 MVA)
DFAX: 8.8%
Pre-project loading: 110.1%
4. Facility: Hampton Rd – Medora 345 kV line
Contingency: Simpson Stuart Tap – Fishkill Plains 345 kV line
Loading: 111.8% (Rating: 1,243 MVA)
DFAX: 8.0%
Pre-project loading: 110.2%
5. Facility: Ritter Park – Stewart Mn Dam 345 kV
Contingency: Medora – P.T.S 345 kV
Loading: 109.0% (Rating: 1,173 MVA)
DFAX: 8.1%
Pre-project loading: 107.3%

6. Facility: Burwell 500/345kV Transformer
Contingency: Allegheny Energy Units 3 4 & 5 345/1kV XFMR
Loading: 107.6% (Rating: 692 MVA)
DFAX: 6.9%
Pre-project loading: 105.1%
7. Facility: Sugar Hill – Allegheny Energy Units 3 4 & 5 345 kV line
Contingency: Piney Crk Project – Hilltop 500 kV
Loading: 107.4% (Rating: 1,163 MVA)
DFAX: 11.1%
Pre-project loading: 105.0%
8. Facility: Woodland Tap – Youngwood 345 kV
Contingency: Southeast Vicksburg – Allegheny Energy Units 3 4 & 5 500 kV
Loading: 106.8% (Rating: 1,494 MVA)
DFAX: 6.3%
Pre-project loading: 105.7%
9. Facility: Fishkill Plains Substation – Dillard-Orange 345 kV Line
Contingency: Tidd – Muskingum River 500 kV
Loading: 105.2% (Rating: 1,399 MVA)
DFAX: 7.0%
Pre-project loading: 103.9%
10. Facility: Allegheny Energy Units 3 4 & 5 500 kV/1 kV Auto
Contingency: Piney Crk Project – Hilltop 500 kV Line
Loading: 101.6% (Rating: 1,114 MVA)
DFAX: 10.7%
Pre-project loading: 99.2%
11. Facility: Allegheny Energy Units 3 4 & 5 345/1kV Transformer
Contingency: Hilltop – Piney Crk Project 500 kV
Loading: 101.6% (Rating: 1,114 MVA)
DFAX: 10.7%
Pre-project loading: 99.2%
12. Facility: Minco – Larson St 230 kV
Contingency: Barrett – Norwich 500 kV Line
Loading: 100.9% (Rating: 1,016 MVA)
DFAX: 5.9%
Pre-project loading: 99.5%
13. Facility: Burwell – P.T.S 345 kV line
Contingency: Allegheny Energy Units 3 4 & 5 345-1 kV Transformer Bank
Loading: 100.8% (Rating: 1,075 MVA)
DFAX: 8.1%
Pre-project loading: 98.9%
14. Facility: Muskogee – Timber Switch 230 kV
Contingency: Swan Vly – Trego 500 kV
Loading: 100.3% (Rating: 417 MVA)
DFAX: 6.7%
Pre-project loading: 96.2%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Dooville – Montpelier Electric Generating Station 230 kV	Rebuild line with higher-capacity conductor	\$25,585,000	47% (\$11,943,000)
Ritter Park – Sugar Hill 345 kV line	Rebuild line with higher-capacity conductor	\$19,437,000	4% (\$810,000)
P.T.S – Medora 345 kV	Rebuild line with higher-capacity conductor	\$30,016,000	4% (\$1,259,000)
Hampton Rd – Medora 345 kV	Rebuild line with higher-capacity conductor	\$31,674,000	4% (\$1,329,000)
Ritter Park – Stewart Mtn Dam 345 kV	Reconductor line	\$8,138,000	4% (\$341,000)
Burwell 500/345kV Transformer	Replace transformer with higher-rated unit	\$26,415,000	4% (\$1,105,000)
Sugar Hill – Allegheny Energy Units 3 4 & 5 345 kV line	Reconductor line	\$8,548,000	4% (\$356,000)
Youngwood – Woodland Tap 345 kV	Reconductor line	\$15,788,000	4% (\$660,000)
Dillard-Oange – Fishkill Plains 345 kV	Reconductor line	\$40,192,000	4% (\$1,743,000)
Allegheny Energy Units 3 4 & 5 500 kV/1 kV Auto	Replace transformer with higher-rated unit	\$28,978,000	4% (\$1,208,000)
Allegheny Energy nits 3 4 & 5 345-1 kV XFMR	Replace transformer with higher-rated unit	\$15,625,000	4% (\$651,000)
Minco – Larson St 230 kV	Reconductor line	\$4,580,000	6% (\$300,000)
P.T.S – Burwell 345 kV	Reconductor line	\$3,940,000	4% (\$165,000)
Muskogee – Timber Switch 230 kV	Reconductor line	\$8,806,000	10% (\$843,000)
	Total allocated to project		\$22,713,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.